Soil Advances 5 (2026) 100113

Contents lists available at ScienceDirect

Soil Advances

journal homepage: www.sciencedirect.com/journal/soil-advances

Mapping 25 years of hydrogel research in soil health: Bibliometric insights into scientific trajectories and global collaboration gaps

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ARTICLE INFO

Keywords:
Superabsorbent polymers
Hydrogels, soil health
Bibliometric analysis
Polyacrylamide

ABSTRACT

Soil degradation and increasing climate variability are intensifying soil moisture limitations in many agricultural regions, particularly in arid and semi-arid environments. Hydrogels (superabsorbent polymers, SAPs) have emerged as promising soil amendments capable of improving soil water retention and enhancing crop performance under water-limited conditions. Despite the growing body of experimental research on hydrogel-soil interactions, the overall intellectual structure and thematic development of this research field remain insufficiently understood. This study provides a comprehensive bibliometric assessment of global research on hydrogels and superabsorbent polymers applied to soil systems during the period 2000–2025. A dataset of 652 peer-reviewed articles indexed in the Scopus database was analysed using the Bibliometrix R package and VOSviewer to examine publication trends, collaboration patterns, and conceptual research themes. Results indicate a strong and accelerating growth in scientific output, particularly after 2015, reflecting increasing global interest in polymer-based soil amendments for water management and climate resilience. Keyword co-occurrence and bibliographic coupling analyses revealed that the literature is primarily structured around four major research domains: soil water management and crop productivity, modification of soil physical properties and water retention, soil structural stabilization and aggregation processes, and hydrogel materials and polymer chemistry. Thematic evolution analysis further indicates a transition from early studies on synthetic polymers toward increasing research attention on biodegradable and bio-based hydrogels derived from natural polymers. However, biological responses of soil systems and socio-economic aspects of hydrogel adoption remain comparatively underexplored. This bibliometric synthesis clarifies the conceptual development of hydrogel research in soil systems and identifies priority areas for future interdisciplinary research on sustainable soil water management.

1. Introduction

Soil degradation continues to pose a major constraint to agricultural productivity, ecosystem stability, and land restoration efforts worldwide. It is estimated that nearly 40% of global land resources are affected by various forms of degradation, including erosion, nutrient depletion, salinization, and declining soil organic matter, particularly in arid and semi-arid regions (Lal, 2015; UNCCD, 2025). These challenges are further intensified under changing climatic conditions, where increasing temperature variability and irregular precipitation patterns exacerbate soil moisture stress and reduce input-use efficiency (Lal, 2015; Montanarella et al., 2016). Among the emerging technological interventions proposed to address soil moisture limitations, hydrogels (superabsorbent polymers, SAPs) have received increasing attention over the past two decades. These materials consist of three-dimensional cross-linked hydrophilic polymer networks capable of absorbing and retaining water many times their own weight and gradually releasing it under moisture-deficit conditions. When incorporated into soil, hydrogels function as temporary water reservoirs that modify soil-water dynamics and influence soil physical, chemical, and biological processes that collectively determine soil functioning (Ahmed, 2015; Neethu et al., 2018). In polymer science, superabsorbent polymers represent a specific class of hydrogels characterized by exceptionally high water absorption capacity; however, in soil and agricultural literature these two terms are

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https://doi.org/10.1016/j.soilad.2026.100113
Received 3 February 2026; Received in revised form 1 April 2026; Accepted 2 April 2026
Available online 7 April 2026
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often used interchangeably because most hydrogels investigated as soil amendments function as superabsorbent materials.

Early studies on SAP applications in soil systems were largely based on synthetic polymers, particularly polyacrylamide and polyacrylate-based materials, which demonstrated improvements in soil water retention, aggregate stability, infiltration behaviour, and erosion control (Ma and Wen, 2020). However, concerns related to cost, environmental persistence, and potential ecological risks have stimulated increasing interest in biodegradable and bio-based hydrogels derived from natural polymers such as starch, cellulose, and chitosan (Behera and Mahanwar, 2020; Thombare et al., 2018). This transition reflects a broader effort to align hydrogel research with sustainable soil management and land restoration strategies.

Despite the expanding body of experimental literature, research on superabsorbent polymers in soil systems remains fragmented across several disciplines, including polymer science, soil physics, agronomy, and environmental engineering (Yang et al., 2021a; Behera and Mahanwar, 2020). Much of the existing work emphasizes short-term laboratory or greenhouse experiments focusing on soil physical properties such as water retention and soil moisture dynamics (Krasnopeeva et al., 2022; Malik et al., 2022), while comparatively fewer studies investigate long-term field performance, soil biological responses, or system-level soil health outcomes (Adjuik et al., 2023). Furthermore, the geographical distribution of research activity appears uneven, with a concentration of publications originating from a limited number of countries despite the global relevance of hydrogel technologies for water-limited and degraded landscapes (Yang et al., 2021a).

In this context, bibliometric analysis provides a systematic framework for examining how scientific knowledge within a research field develops over time. By analysing citation networks, keyword co-occurrence patterns, and conceptual clustering of publications, bibliometric approaches enable quantitative mapping of the intellectual structure and thematic evolution of scientific literature. Unlike narrative or traditional review methods, bibliometric analyses allow synthesis of large bodies of research in an objective and reproducible manner while identifying dominant research domains, emerging themes, and knowledge gaps within interdisciplinary fields (Aria and Cuccurullo, 2017; Kumar, 2025).

Although numerous reviews have examined hydrogel synthesis, material properties, and agronomic applications, these studies primarily summarize experimental findings rather than analysing how the research field itself has evolved. Consequently, the broader intellectual structure, thematic development, and emerging research priorities in hydrogel-soil studies remain insufficiently characterized. Understanding these patterns is important for identifying future research directions and for aligning polymer-based soil amendment research with global challenges related to soil degradation and water scarcity.

Therefore, the present study conducts a bibliometric analysis of peer-reviewed literature on hydrogels applied to soil systems between 2000 and 2025. Specifically, this study addresses three key questions: (i) how has research on hydrogel-based soil amendments evolved over time; (ii) what conceptual domains structure this interdisciplinary research field; and (iii) which emerging themes and knowledge gaps require further investigation? The study examines temporal trends in scientific production, global patterns of research activity and collaboration, and the conceptual organization of research themes through keyword and citation network analyses. By identifying dominant research domains and underexplored areas, this analysis aims to clarify the intellectual development of hydrogel-soil research and highlight priority directions for future investigations addressing soil water management and sustainable land management challenges.

2. Materials and methods

2.1. Data source, search strategy, and screening protocol

A comprehensive bibliometric dataset was constructed using the Scopus database, which was selected due to its extensive multidisciplinary coverage, standardized indexing system, and reliable citation metadata. Compared with alternative bibliographic databases, Scopus provides broader representation of environmental, agricultural, and earth system sciences, making it particularly suitable for bibliometric analyses in soil research (Pranckutė, 2021; Fragomeli et al., 2024; Mahanta et al., 2025). The literature search was conducted on 16 January 2026 and the analysis covered publications indexed between 2000 and 2025. The starting year 2000 was selected because research on superabsorbent polymers and hydrogels in soil and agricultural systems expanded substantially after the early 2000s with the development of modern polymer-based soil amendments, whereas earlier studies were relatively limited and primarily focused on industrial applications (Ahmed, 2015; Ma and Wen, 2020). The search query included both "hydrogel" and "superabsorbent polymer" terminology to capture studies describing functionally similar water-absorbing polymer amendments used in soil systems. These soil-related descriptors were selected because they represent commonly reported indicators of soil physical, hydrological, and biological responses to polymer-based soil amendments.

The search strategy employed a structured Boolean query applied to titles, abstracts, and author keywords. Polymer-related terms were combined with soil-relevant descriptors to ensure thematic specificity and to minimize retrieval of studies unrelated to soil processes. The final search query consisted of polymer descriptors including superabsorbent polymer, hydrogel, polyacrylamide, biopolymer, and chitosan hydrogel, in combination with soil-related terms such as soil moisture, aggregation, microbial biomass, and bulk density. This approach ensured that retrieved records explicitly addressed soil physical, hydrological, or biological responses to polymer-based amendments. The exact keyword used was TITLE-ABS-KEY ("superabsorbent polymer*" OR "hydrogel*" OR "polyacrylamide*" OR "biopolymer*" OR "chitosan hydrogel") AND TITLE-ABS-KEY ("soil moisture" OR "aggregation" OR "microbial biomass" OR "bulk density").

2.2. Inclusion criteria

Following the initial retrieval, records were screened based on predefined inclusion criteria to ensure consistency and thematic relevance. Only peer-reviewed research articles published between 2000 and 2025 were retained, while reviews, conference papers, book chapters, notes, and early-access items were excluded. The analysis was restricted to final published articles in academic journals written in English to maintain comparability of citation and keyword metadata. To ensure the disciplinary relevance, the dataset was further limited to studies that explicitly investigated the application or effects of hydrogels or superabsorbent polymers in soil systems. Publications were therefore retained only when the title, abstract, or author keywords indicated a clear focus on soil-related processes such as soil moisture dynamics, soil physical properties, soil-plant interactions, or soil amendment effects.

The dataset was further limited to Scopus subject areas in which soil constitutes the primary system of investigation, including Agricultural and Biological Sciences, Environmental Science, and Earth and Planetary Sciences. Records indexed under multidisciplinary categories were evaluated individually and excluded where soil-related processes were not central to the study objectives.

2.3. Exclusion criteria

Given the extensive use of polyacrylamide and hydrogel materials in biomedical, biochemical, and molecular research, a rigorous exclusion

procedure was applied to prevent thematic contamination of the dataset. Publications focusing on protein chemistry, molecular biology, pharmaceutical formulations, or medical applications were explicitly removed. These included studies related to protein aggregation, polyacrylamide gel electrophoresis, in-vitro experiments, metabolic or genetic analyses, drug delivery systems, and animal or human studies. These exclusion criteria were essential because polymeric materials are widely used in laboratory and clinical contexts that are unrelated to soil structure, soil-water dynamics, aggregation processes, or soil biological functioning. Removing such records ensured that the final dataset accurately reflected soil-focused applications of hydrogels and superabsorbent polymers, rather than analytical or biomedical uses of similar materials.

After applying all inclusion and exclusion criteria, a total of 652 peer-reviewed research articles were retained for bibliometric analysis. The curated dataset represents global research on the use of hydrogels in soil systems, with particular emphasis on soil moisture regulation, water retention, aggregation, soil physical properties, organic matter dynamics, and soil-plant interactions. This dataset formed the foundation for all subsequent descriptive, network-based, and thematic analyses.

2.4. Bibliometric and network analyses

Quantitative bibliometric analyses, including descriptive statistics, keyword co-occurrence mapping, and thematic evaluation, were conducted using the Bibliometrix R package through its Biblioshiny web interface. This platform enabled standardized computation of publication trends, citation indicators, and thematic structures (Aria and Cuccurullo, 2017; Kumar, 2025). Network visualizations and clustering analyses were performed using VOSviewer (version 1.6), which was employed to construct keyword co-occurrence networks (Van Eck and Waltman, 2010). Clustering was carried out using the Louvain community detection algorithm, and cluster interpretation was based on author keywords and normalized citation indicators. To explore the conceptual structure of the research field, Multiple Correspondence Analysis was applied to keyword co-occurrence data, allowing identification of dominant conceptual dimensions and thematic groupings within the literature. For keyword co-occurrence analysis, a minimum occurrence threshold of 5 was applied to author keywords to ensure network stability and interpretability, resulting in the inclusion of approximately 90 keywords in the network.

3. Results

3.1. Overview of bibliometric dataset

The bibliometric dataset comprised 652 peer-reviewed articles published between 2000 and 2025 and indexed in the Scopus database (Table 1). These publications were distributed across 295 sources. A total of 2674 authors contributed to the dataset, with an average of 5.42 co-authors per article. The dataset included 2316 author keywords and 4314 Keywords Plus terms. Detailed bibliometric indicators describing author productivity, institutional contributions, leading publication sources, and collaboration networks are provided in the Supplementary Material (Tables S1-S4 and Figures S1-S2). These additional analyses describe the broader structure of the research community but are not central to the conceptual and thematic analyses that form the focus of this study.

3.2. Annual scientific production and average citations per year

The annual scientific production related to hydrogels in soil systems shows a clear temporal increase during the study period (2000–2025) (Fig. 1). During the early phase of the dataset (2000–2006), research output remained limited, with only a few publications appearing each year. Publication activity increased gradually after 2007 and remained relatively stable between 2008 and 2014, when annual outputs generally ranged between 10 and 13 articles. A more pronounced increase in publication activity occurred after 2015. Annual output rose steadily during this period and increased sharply after 2019. The highest levels of scientific production were observed in the final years of the dataset, with 89 publications in 2024 and 110 publications in 2025.

Average citations per article were higher for earlier publications and declined for more recent years (Fig. 1), with early articles showing comparatively higher citation values than recent publications.

Table 1
Main information about the data.

Main information about data	
Timespan	2000:2025
Sources (Journals, Books, etc)	295
Documents	652
Annual Growth Rat%	20.69
Document Average Age	6.69
Average citations per doc	27.7
References	5786
DOCUMENT CONTENTS	
Keywords Plus (ID)	4314
Author's Keywords (DE)	2316
AUTHORS	
Authors	2674
Authors of single-authored docs	7
AUTHORS COLLABORATION	
Single-authored docs	8
Co-Authors per Doc	5.42
International co-authorships %	23.62
DOCUMENT TYPES	
article	652

3.3. Global distribution of research activity and collaboration patterns

The geographical distribution of research activity indicates that scientific output is concentrated in a limited number of countries. China (578 publications), the United States (161), and India (131) were the leading contributors (Fig. 2), followed by Brazil (54), Iran (61), Australia (61), Germany (45), and Italy (42). International collaboration networks indicate a globally connected research structure (Fig. 3). China and the United States occupy central positions within the network and maintain collaborations with multiple countries. India also shows international collaboration, particularly with the United States, the United Kingdom, and Saudi Arabia, while Brazil and Australia exhibit collaborations across multiple regions. Detailed country-level publication statistics and collaboration metrics are provided in Supplementary Table S4.

3.4. Bibliographic coupling and research domains

Bibliographic coupling analysis was performed at the document level to identify groups of publications that share similar reference patterns (Fig. 4). The resulting network revealed several clusters of documents with related citation linkages, which can be interpreted as indicating areas of closely connected research activity within hydrogel studies in soil systems. The largest cluster was associated with keywords such as hydrogel, soil moisture, and superabsorbent polymer, suggesting a concentration of studies related to soil moisture regulation and drought-related applications. This cluster also exhibited relatively high connectivity within the network, indicating strong linkage among publications related to water retention and soil water availability. In contrast, a second cluster centred around polyacrylamide (PAM) and soil erosion is associated with studies linked to soil stabilization and erosion processes, with several studies examining the role of polymer amendments in aggregation and erosion dynamics. Keywords related to aggregation, aggregate stability, and soil structure appear grouped within another cluster, reflecting a set of studies addressing soil physical condition and

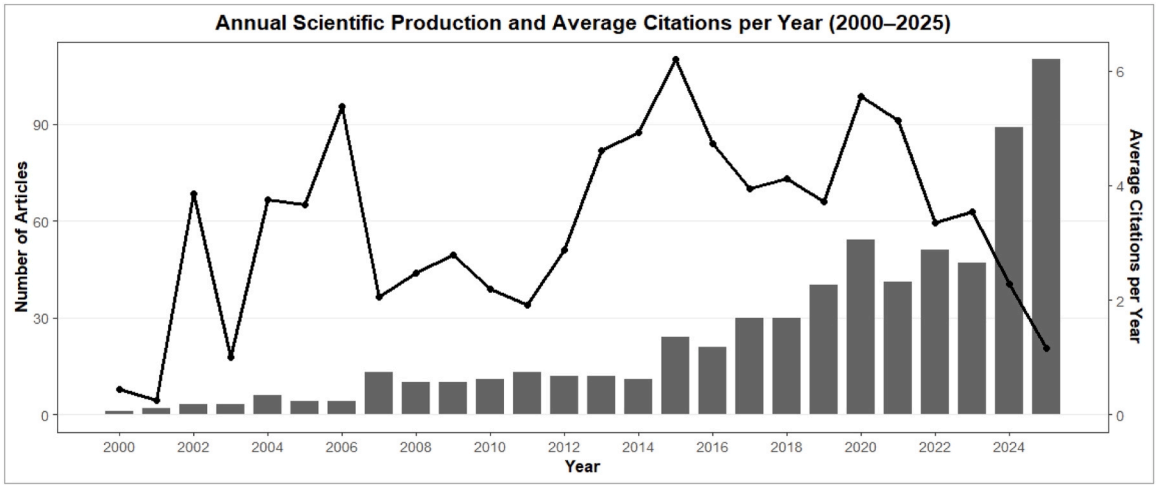


Fig. 1. Annual scientific publication trend related to hydrogels and average citations per year from 2000 to 2025.

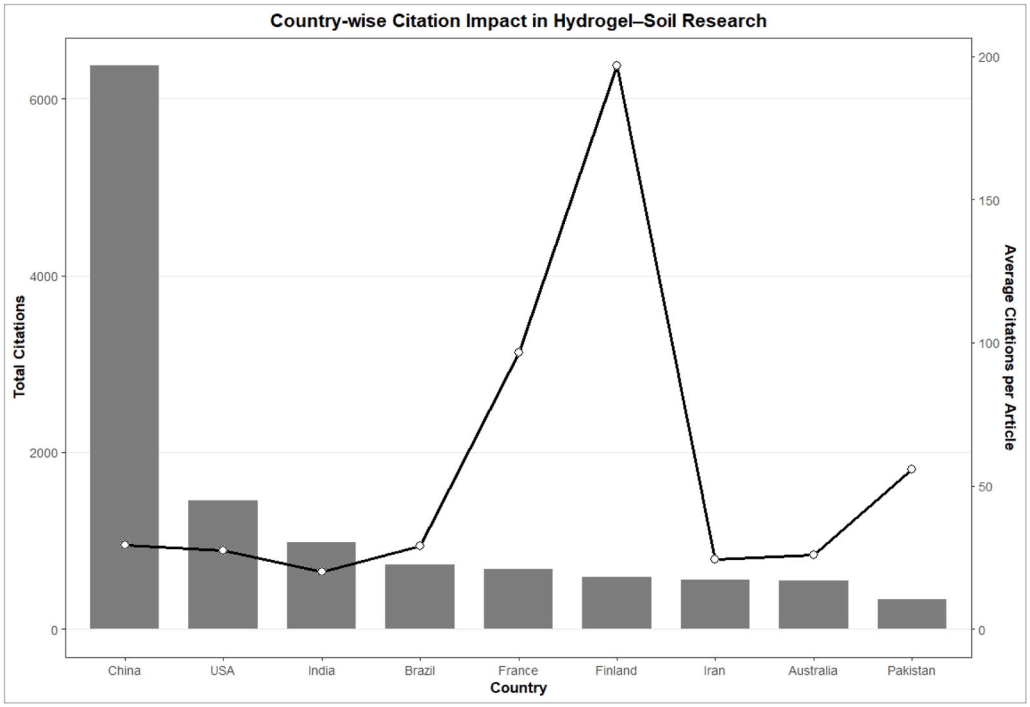


Fig. 2. Country-wise citation impact in hydrogels research.

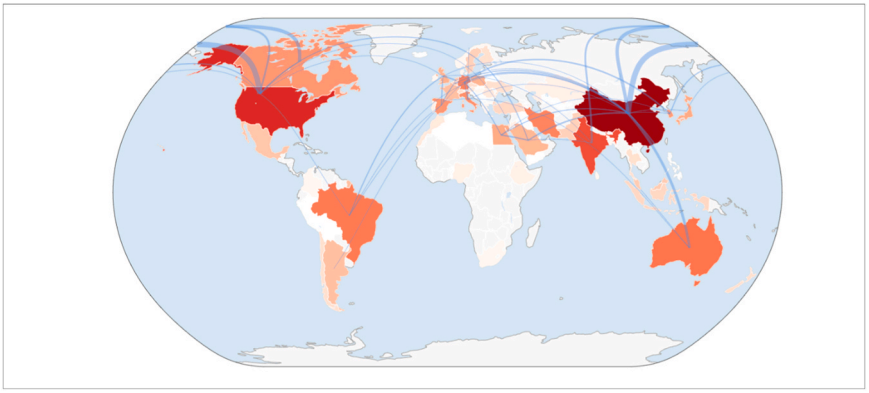


Fig. 3. International collaboration network among countries on research related to hydrogels.

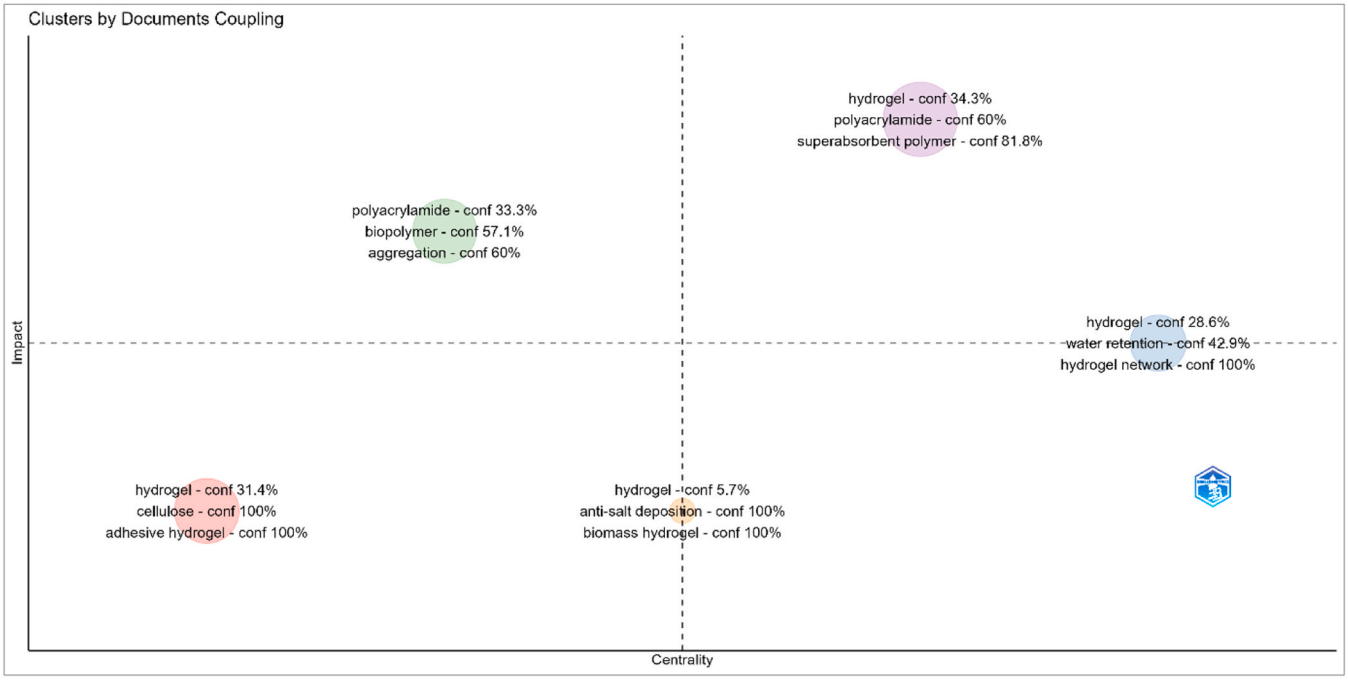


Fig. 4. Conceptual clusters identified through document-level bibliographic coupling showing the centrality and impact of major research domains in hydrogel studies in soil systems.

structural properties influenced by polymer-based amendments. Additional clusters reflect more recent and specialized directions, including biopolymers, biochar, carbon-related processes, as well as biodegradable hydrogel materials. Smaller groupings associated with adsorption and contaminant immobilization further indicate environmental applications of polymer-based soil amendments.

3.5. *Keyword co-occurrence and thematic structure of the literature*

Keyword co-occurrence analysis was performed to identify major

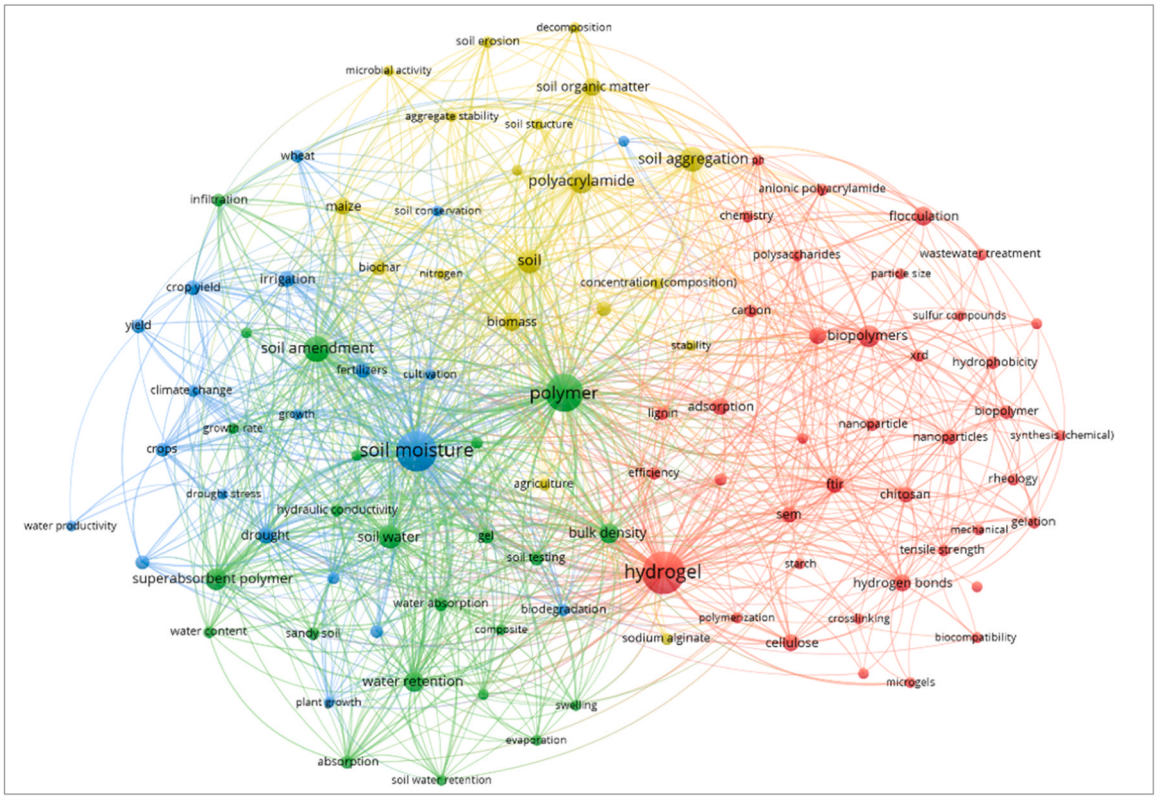


Fig. 5. Keyword co-occurrence network illustrating the conceptual structure of hydrogel research in soil systems. Node size represents keyword frequency, link thickness indicates co-occurrence strength, and colors represent major thematic clusters identified in the literature.

research themes associated with hydrogel applications in soil systems (Fig. 5). The network revealed several interconnected clusters reflecting distinct but related areas of research within the literature.

The most prominent cluster is centred on soil water management and crop productivity, with keywords such as soil moisture, drought, irrigation, yield, and crop growth. This cluster reflects a concentration of studies examining the role of hydrogels in enhancing soil water availability and plant performance under water-limited conditions. Another cluster brings together terms related to soil physical properties and water retention, including superabsorbent polymer, soil amendment, water retention, soil water, and bulk density. These keywords indicate a focus on how hydrogel applications influence soil physical characteristics and water-holding capacity. Keywords related to soil aggregation, soil organic matter, polyacrylamide, and microbial activity appear grouped within a third cluster, linking studies concerned with soil structural stability and biogeochemical interactions. In contrast, a separate cluster is defined by terms such as hydrogel, biopolymers, adsorption, chitosan, and nanoparticles, reflecting research related to polymer materials, synthesis approaches, and associated environmental applications.

3.6. Thematic evolution of research on hydrogels and soil systems

The thematic evolution analysis illustrates how research topics related to hydrogels in soil systems have developed across four time periods (2000–2007, 2008–2014, 2015–2021, and 2022–2025) (Fig. 6). Rather than isolated shifts, the evolution pattern indicates a progressive expansion of research focus from polymer-assisted soil structural modification toward more diversified material and application-oriented themes.

During the early phase (2000–2007), research was concentrated on a limited set of topics, particularly polyacrylamide, chitosan, and soil aggregation, reflecting initial efforts to understand polymer-based soil amendments and their influence on soil structure. These early themes persisted into subsequent periods, indicating continuity in core research areas.

Between 2008 and 2014, the thematic structure expanded to include broader soil amendment concepts such as superabsorbent polymers and hydrophilic materials. This phase marks a transition from narrowly focused material studies toward more applied perspectives involving soil modification and water management.

The period from 2015 to 2021 shows a clear diversification of research themes, with the emergence of biopolymers, cellulose-based materials, and rheological studies. This shift suggests increasing interest in alternative polymer materials and more complex functional properties, reflecting a broadening of the research scope beyond conventional synthetic polymers.

In the most recent period (2022–2025), the thematic structure indicates continued development of both established and emerging topics. Core themes such as hydrogel, superabsorbent polymer, and soil aggregation remain prominent, while newer materials such as sodium alginate and lignin appear alongside biopolymer-related research. This pattern suggests a sustained research foundation combined with ongoing diversification toward novel and potentially more sustainable polymer materials.

Building on the thematic evolution patterns described above, further analysis of keyword frequency, conceptual relationships, and thematic importance was conducted to characterize the intellectual structure of hydrogel research in soil systems.

3.7. Conceptual structure of research themes in hydrogel-soil studies

3.7.1. Most frequent research keywords

The analysis of author keywords revealed the most frequently occurring research terms associated with hydrogels in soil systems (Table 2). The keyword hydrogel appeared most frequently with 182 occurrences, followed by soil moisture (159), polymer (136), soil amendment (65), and soil aggregation (64). Other frequently occurring terms included polyacrylamide (52), superabsorbent polymer (48), soil water (43), and biopolymers (41).

Several keywords related to soil physical properties were also prominent, including bulk density, water retention, soil organic matter, and soil water. In addition, terms related to soil-polymer interaction processes such as flocculation, adsorption, and agglomeration appeared frequently within the dataset. Keywords related to analytical characterization methods, including FTIR, SEM, and XRD, also appeared in the literature, indicating the role of material characterization techniques in hydrogel research.

The presence of agronomic and crop-related keywords such as maize (35 occurrences), wheat (19), crop yield (20), irrigation (21), and drought (23) indicate that agronomic applications and water management constitute a prominent component of the keyword landscape in the

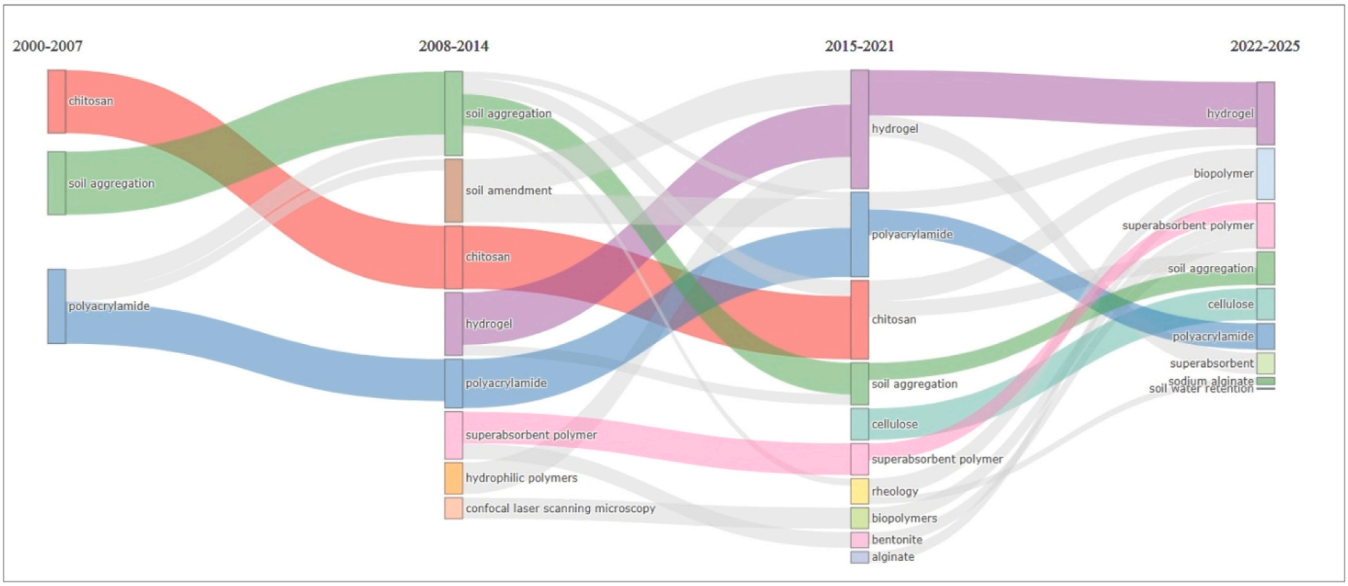


Fig. 6. Thematic evolution of research topics related to hydrogels in soil systems across four time periods (2000–2007, 2008–2014, 2015–2021, and 2022–2025). The flows illustrate continuity and transformation of dominant keywords over time.

Table 2
Most frequent author keywords in research on hydrogels and superabsorbent polymers in soil systems.

Keyword	Occurrences
hydrogel	182
soil moisture	159
polymer	136
soil amendment	65
soil aggregation	64
polyacrylamide	52
soil	50
superabsorbent polymer	48
soil water	43
biopolymers	41
bulk density	36
maize	35
water retention	35
soil organic matter	32
flocculation	29
biomass	28
FTIR	27
adsorption	26
agglomeration	26
cellulose	26

dataset.

3.7.2. Conceptual structure of research themes based on factorial analysis

The conceptual relationships among research themes were examined using multiple correspondence analysis (MCA) based on author keywords. The factorial map (Fig. 7) represents the spatial distribution of keywords according to two principal dimensions (Dim1 = 55.38% and Dim2 = 14.8%), which together explain a substantial proportion of the variance in the dataset.

Keywords positioned along the positive side of the first dimension include soil moisture, superabsorbent polymer, water retention, water absorption, irrigation, and crop yield, which may reflect a gradient related to soil water management and agricultural productivity. In contrast, several keywords associated with material properties and physicochemical characterization, such as hydrogen bonds, cellulose, SEM, nanoparticles, and FTIR, appear distributed along the negative region of the first dimension.

The second dimension appears to be associated with soil structural and amendment-related processes. Keywords such as polyacrylamide, soil aggregation, soil organic matter, and biochar are located toward the positive side of this dimension, suggesting a linkage with soil structural modification and amendment-based soil management.

Other keywords related to agronomic performance, including maize, wheat, crop yield, and irrigation, appear clustered in the upper-right portion of the factorial map, which may indicate their co-occurrence in studies focusing on crop responses to hydrogel-based soil amendments.

3.7.3. Thematic structure of research topics

The thematic map based on centrality and density provides an overview of the relative positioning of major research themes within the field (Fig. 8). The map divides themes into four quadrants representing motor themes, niche themes, basic themes, and emerging or declining themes.

Keywords related to soil moisture, polymer, and soil amendment appear as relatively central themes within the thematic map, suggesting stronger connectivity with other research topics in the literature. These themes may represent a central area of focus linking soil physical processes with polymer-based soil management strategies.

Themes related to soil aggregation, polyacrylamide, and soil appear within the niche theme quadrant, suggesting that although these topics are relatively well developed, they are less connected with other thematic areas within the dataset. In contrast, themes associated with hydrogel, biopolymers, and flocculation appear within the emerging or developing theme quadrant, reflecting comparatively lower density and centrality within the thematic structure.

The thematic map suggests that research on hydrogels in soil systems is oriented around soil water management and polymer-based soil amendments, while several specialized research themes continue to develop within the broader literature. It should be noted that thematic positioning is influenced by keyword selection and threshold parameters used in the analysis.

4. Discussion

This bibliometric analysis provides a comprehensive overview of global research on hydrogels in soil systems over the period 2000–2025. The results highlight the rapid expansion of scientific activity, the

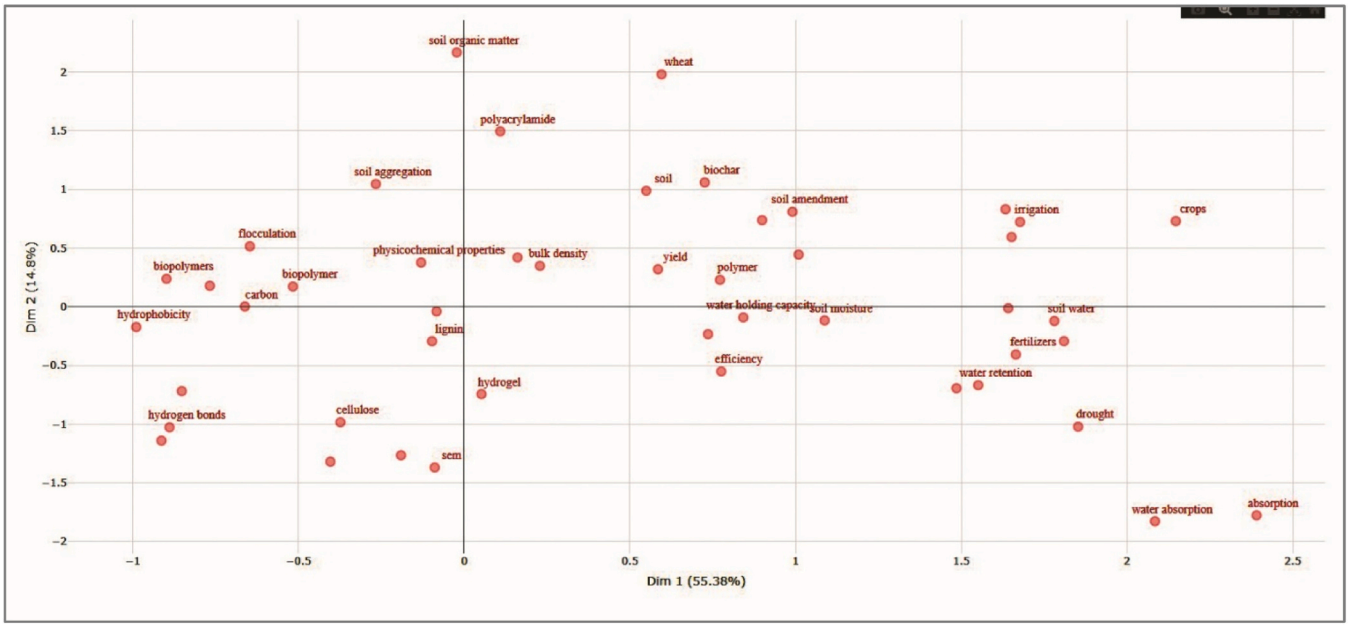


Fig. 7. Conceptual structure map of research themes related to hydrogels in soil systems based on multiple correspondence analysis (MCA).

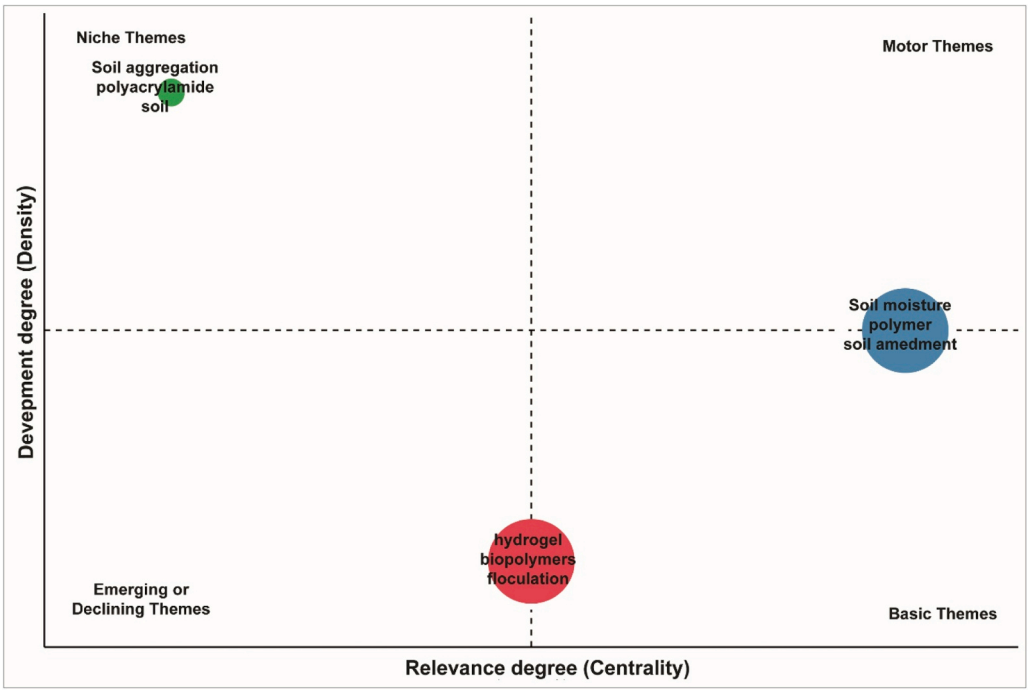


Fig. 8. Thematic map showing the centrality and development of research themes related to hydrogels. Themes are categorized as motor themes, niche themes, basic themes, and emerging or declining themes based on their centrality and density values.

conceptual organization of research themes, and the evolving structure of knowledge in this interdisciplinary field. The steady increase in annual scientific production observed in this study suggests increasing research attention toward polymer-based soil amendments, particularly in the context of water scarcity, soil degradation, and climate-related stress in agricultural systems (Lal, 2015; Montanarella et al., 2016). The pronounced acceleration of publication activity after 2015, and particularly after 2019, indicates growing research focus on hydrogels and SAPs in relation to climate-resilient soil management strategies.

The sustained growth of the literature is consistent with broader trends observed for emerging soil amendment technologies, where research activity increases as applications diversify and new environmental challenges stimulate innovation (Adhikari and Hartemink, 2016). In the case of hydrogels, the scientific focus appears to have expanded from early materials-based investigations toward studies addressing soil physical processes, plant–soil interactions, and agricultural water management.

Bibliographic coupling and keyword co-occurrence analyses indicate that research on hydrogels in soil systems is organized around several interconnected areas of study. The most prominent cluster identified in this study is associated with soil water dynamics, including themes related to soil moisture, water retention, drought mitigation, irrigation, and crop productivity. The prominence of these keywords in the bibliometric analysis is consistent with experimental studies describing the role of hydrogels in regulating soil water availability. Superabsorbent polymers consist of three-dimensional hydrophilic polymer networks containing functional groups such as carboxyl, amide, hydroxyl, and sulfonic groups that enable the absorption and retention of large quantities of water (Qureshi et al., 2020; Saha et al., 2020). Once incorporated into soil, swollen polymer particles can act as localized water reservoirs that absorb excess water during rainfall or irrigation and gradually release it as soil moisture declines.

Experimental studies have reported that SAP amendments can improve soil water availability and plant performance under water-limited conditions (Akhter et al., 2004; Bai et al., 2010). Hydrogels may reduce evaporative losses and prolong soil moisture availability, thereby delaying the onset of plant water stress. However, it should be noted that the present bibliometric analysis does not directly evaluate these mechanisms but rather reflects their prominence in the literature. The strong association between hydrogels and keywords such as soil moisture, water retention, drought, irrigation, and crop yield therefore indicates the centrality of these topics within the research field.

In addition to regulating soil water availability, hydrogels can significantly modify soil physical properties, as reported in experimental studies. Several studies have reported reductions in soil bulk density and increase in soil porosity and plant-available water following SAP application, particularly in coarse-textured soils (Abrisham et al., 2018). The repeated swelling and shrinking of polymer particles can create additional pore spaces within the soil matrix, thereby improving soil aeration and water storage capacity (Parvathy et al., 2014).

Hydrogels can also influence soil hydraulic processes such as infiltration and saturated hydraulic conductivity. Moderate SAP application rates may increase infiltration and cumulative water storage, whereas excessive polymer doses may partially block soil pores and reduce hydraulic conductivity due to swelling of polymer particles within soil pore networks (Albalasmeh et al., 2022). The water absorption and swelling behaviour of superabsorbent polymers is controlled by the expansion of hydrophilic functional groups within cross-linked polymer networks, which generate osmotic pressure and facilitate large water uptake and gradual release. Similar swelling and water-retention mechanisms have also been documented in studies examining SAP behaviour in cementitious and other material systems, where polymer hydration and expansion influence the microstructure and moisture dynamics of surrounding matrices (Sujitha and Ramesh, 2023; Xavier, 2023). These mechanistic insights are derived from experimental studies reported in the literature. In the context of the present study, the bibliometric patterns do not directly demonstrate these processes but indicate the prominence of related topics such as bulk density, water retention, soil aggregation, and infiltration within the research field.

Another prominent research domain identified through bibliographic coupling is related to soil structural stabilization and erosion control. Polymer-based soil conditioners, particularly polyacrylamide-based materials, have long been studied for their ability to enhance soil aggregation and reduce erosion. Charged functional groups present

in polymer chains promote flocculation and aggregation of soil particles, thereby increasing aggregate stability and resistance to soil detachment (Bai et al., 2010). Long-term experiments have demonstrated that SAP application can increase the proportion of macro-aggregates and improve mean weight diameter of soil aggregates, indicating improved soil structural stability (Yang et al., 2021b). The continued presence of polyacrylamide and soil aggregation as frequently occurring keywords across multiple thematic clusters indicates that polymer-assisted soil stabilization remains a prominent theme within the literature. However, this bibliometric prominence reflects research attention rather than the relative scientific importance or practical effectiveness of these approaches.

Conceptual structure analysis further highlights the interdisciplinary nature of hydrogel research in soil systems. In addition to soil physical and agronomic keywords, the literature contains numerous terms associated with polymer chemistry and material characterization, including FTIR, SEM, hydrogen bonding, and nanoparticle-based materials. These keywords reflect the interaction between soil science and material engineering within this research field. The co-occurrence of these keywords in the literature is consistent with experimental studies indicating that hydrogel-based soil amendments involve both soil management processes and polymer synthesis, cross-linking chemistry, and material characterization techniques (Ahmed, 2015; Rizwan et al., 2021; Kothari et al., 2022). Such interdisciplinary linkages are associated with the diversification of hydrogel materials and the development of specialized polymer formulations for agricultural and environmental applications

The thematic evolution analysis suggests that the intellectual structure of hydrogel research has progressively diversified over time. Early research in the 2000–2007 period focused primarily on synthetic polymers such as polyacrylamide and their role in soil stabilization and erosion control (Sojka and Entry, 2000; Mamedov et al., 2010). During the following decade, research attention appears to have expanded toward soil water management and plant-soil interactions. More recently, the literature shows increasing emphasis on biodegradable and bio-based hydrogel materials derived from natural polymers such as starch, cellulose, and chitosan (Ahmed, 2015; Thombare et al., 2018; Adjuik et al., 2023). These patterns may reflect growing interest in environmentally compatible polymer materials; however, keyword trends alone do not establish a definitive field-wide conceptual transition.

Environmental sustainability has therefore emerged as a visible dimension within SAP research. Synthetic polyacrylate-based polymers can exhibit relatively slow degradation rates in soil environments. Laboratory studies indicate that mineralization of such polymers may remain below 10% over several months under natural conditions (Stahl et al., 2000). These findings have contributed to increasing interest in biodegradable hydrogel formulations derived from natural polymers such as starch and cellulose. Biopolymer-based hydrogels have been reported to exhibit significantly higher biodegradation rates when exposed to microbial activity in soil systems (Sarkar et al., 2021). The increasing occurrence of keywords such as biopolymers, cellulose, and biodegradable hydrogels in recent literature suggests a shift in research emphasis toward environmentally sustainable materials, rather than confirming a complete conceptual transformation of the field.

Despite the strong focus on soil physical and hydrological processes, the bibliometric patterns identified in this study suggest that biological aspects of soil functioning are less prominently represented in the keyword landscape. Experimental studies have reported increase in microbial biomass, microbial respiration, and bacterial gene abundance following hydrogel application due to improved soil moisture conditions (Khodadadi Dehkordi, 2018). Hydrogel amendments can also influence microbial community composition and increase bacterial and fungal populations in the rhizosphere (Parvathy et al., 2014; Wang et al., 2020). However, such biological responses appear less frequently among dominant research keywords compared with soil physical indicators. This pattern suggests lower visibility of microbial and biochemical processes in the indexed literature, rather than confirming a definitive research gap, although it may indicate areas where further investigation could be beneficial.

The geographical distribution of research activity identified in this study reveals a concentration of scientific output within a limited number of countries. China, the United States, and India collectively account for a substantial proportion of global publications in this field. These patterns are likely influenced by differences in research capacity, funding availability, and publication output across countries. Regions experiencing severe soil degradation and water scarcity, including parts of Africa and Latin America, appear less prominently represented in the SAP research landscape (Montanarella et al., 2016). However, this distribution should be interpreted with caution, as bibliometric patterns primarily reflect research activity and publication trends which might depend on funding rather than the global distribution of environmental challenges.

International collaboration networks derived from the bibliometric analysis indicate that research partnerships are concentrated among a relatively small group of highly productive countries. Although international collaboration has increased over time, greater participation of research institutions from underrepresented regions may enhance the broader applicability of hydrogel-based soil management technologies across diverse environmental conditions (Guilherme et al., 2015).

Finally, the thematic structure of the literature suggests that socio-economic dimensions of hydrogel application are less prominently represented in the keyword landscape. Keywords related to economic feasibility, technology adoption, and cost–benefit analysis appear relatively infrequently among dominant research themes. However, this pattern should be interpreted with caution, as the lower visibility of such keywords may reflect differences in terminology, indexing practices, or database coverage rather than the absence of socio-economic research in practice. Nevertheless, economic considerations remain important for the practical adoption of hydrogel technologies, particularly among smallholder farming systems in water-limited regions (Behera and Mahanwar, 2020; Niu et al., 2025). Future studies integrating agronomic performance, environmental sustainability, and economic feasibility could help strengthen the linkage between experimental research and field-scale implementation.

4.1. *Future research directions*

The bibliometric patterns identified in this study suggest several directions for future research. Based on the observed lower visibility of biological and microbial-related keywords, greater emphasis could be placed on investigating the biological and biochemical responses of soil systems to hydrogel amendments, including microbial community dynamics, enzyme activities, and nutrient cycling processes associated with SAP application.

Given the prominence of keywords related to synthetic polymers and soil physical processes, long-term field experiments would be valuable for evaluating the environmental persistence and degradation behaviour of these materials in soil environments. Understanding their long-term fate is important for assessing environmental sustainability (Yang et al., 2024).

The increasing occurrence of keywords associated with biopolymers and biodegradable materials suggests growing research interest in alternative formulations. Further work on biodegradable hydrogel materials derived from natural polymers such as starch, cellulose, and chitosan could help advance environmentally compatible soil amendments.

The co-occurrence of soil science and material characterization-related keywords indicates interdisciplinary linkages within the field. Strengthening integration across soil science, polymer chemistry, material engineering, and agronomy may support the development of advanced hydrogel formulations tailored to specific soil and climatic conditions.

Finally, the relatively low frequency of socio-economic keywords suggests limited visibility of these aspects in the indexed literature. Greater integration of socio-economic and management perspectives, including economic feasibility, farmer adoption, and large-scale field implementation, could improve the applicability of hydrogel technologies in agricultural systems. It should be noted that these directions are inferred from bibliometric patterns and are intended as indicative priorities rather than conclusions directly derived from the analysis.

4.2. Study limitations

Although this study provides a comprehensive bibliometric assessment of hydrogel research in soil systems, several limitations should be acknowledged. First, the analysis was based exclusively on publications indexed in the Scopus database. Although Scopus provides extensive coverage of international scientific literature, relevant studies published in other databases or non-indexed sources may not have been included. Second, bibliometric analyses rely on publication metadata such as author keywords, citation counts, and collaboration networks. These indicators reflect patterns of scientific activity but do not directly measure the quality or experimental outcomes of individual studies.

Third, keyword-based analyses depend on the consistency of terminology used by authors. Variations in keyword usage may influence the representation of specific research themes within bibliometric networks. Finally, bibliometric indicators primarily describe patterns of scientific communication rather than the practical effectiveness of hydrogel technologies in agricultural systems. These limitations should be considered when interpreting the broader implications discussed in this study, particularly where bibliometric patterns are used to infer research trends and thematic priorities. Future research integrating bibliometric analysis with systematic experimental reviews or meta-analyses could provide a more comprehensive understanding of the scientific and practical impacts of hydrogel-based soil amendments.

5. Conclusions

This study provides a comprehensive bibliometric assessment of global research on hydrogels applied to soil systems over the period 2000–2025. The analysis reveals a rapidly expanding and increasingly interdisciplinary research field that integrates soil science, agronomy, polymer chemistry, and environmental science. The strong growth in publication output observed during the past decade reflects increasing scientific and practical interest in polymer-based soil amendments as potential tools for improving soil water management and enhancing agricultural resilience under climate variability.

Bibliometric network analyses indicate that research in this field is primarily structured around several interconnected domains. The most prominent research themes focus on soil moisture regulation, water retention, drought mitigation, and crop productivity, highlighting the central role of hydrogels in improving soil water availability. Additional important research domains include soil structural stabilization, polyacrylamide-based soil conditioning, and polymer-assisted soil aggregation processes. These themes reflect the capacity of hydrogel materials to modify soil physical properties such as bulk density, porosity, infiltration, and aggregate stability.

Conceptual and thematic analyses further demonstrate that hydrogel research has evolved from early studies focused on synthetic polymer materials toward a broader scientific framework addressing soil-plant interactions, soil structural processes, and environmental sustainability. Recent research trends show increasing interest in biodegradable and bio-based hydrogel materials derived from natural polymers such as starch, cellulose, and chitosan. This shift reflects growing concern regarding the environmental persistence of synthetic polymers and the need for sustainable soil amendment technologies.

Despite the rapid growth of the literature, several research gaps remain. Biological responses of soil systems, including microbial activity and nutrient cycling processes, are comparatively less explored than soil physical properties. In addition, socio-economic aspects of hydrogel adoption in agricultural systems remain weakly represented in the current literature. Addressing these gaps through interdisciplinary research and long-term field evaluations will be essential for advancing the scientific understanding and practical application of hydrogel technologies in soil management.

CRediT authorship contribution statement

Atahar Perwez: Writing – review & editing, Visualization, Validation, Supervision. **Pawan Parihar:** Formal analysis, Data curation. **Sharat Kothari:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Nymisha Alapati:** Writing – review & editing, Writing – original draft, Supervision. **Manorath Sen:** Visualization, Validation, Methodology, Investigation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used an AI-assisted language tool (ChatGPT, OpenAI) solely for language refinement and editorial support. All scientific content, analysis, interpretation, and conclusions were developed, reviewed, and approved by the authors, who take full responsibility for the content of the published article.

Funding

This research received no external funding.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.soilad.2026.100113.

Data availability

Data will be made available on request.

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